5 A coil is placed in a region of magnetic field which is directed into the plane of the paper as shown in Fig. 5.1.

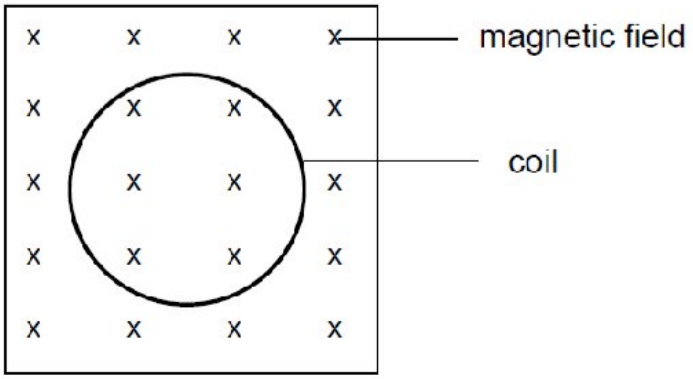


Fig. 5.1

(a) Define *magnetic flux*.

.....

.....

.....[1]

(b) **(i)** The magnitude of the flux density of the magnetic field is decreasing with respect to time. Explain, using Faraday's law, why there is an induced *current* in the coil.

.....

.....

.....[2]

(ii) Draw, on Fig. 5.1, the direction of the induced current. Explain, using Lenz's law, why the induced current is in this direction.

.....

.....

.....[2]

(c) A magnet of length x and magnetic flux density B is dropped vertically from rest with the north pole facing downwards. It is dropped from a reference line which is at a vertical height h above the top of a *long* solenoid of length L and uniform cross-sectional area as shown in Fig. 5.2. The length of the magnet x is much smaller than the length of the solenoid L.

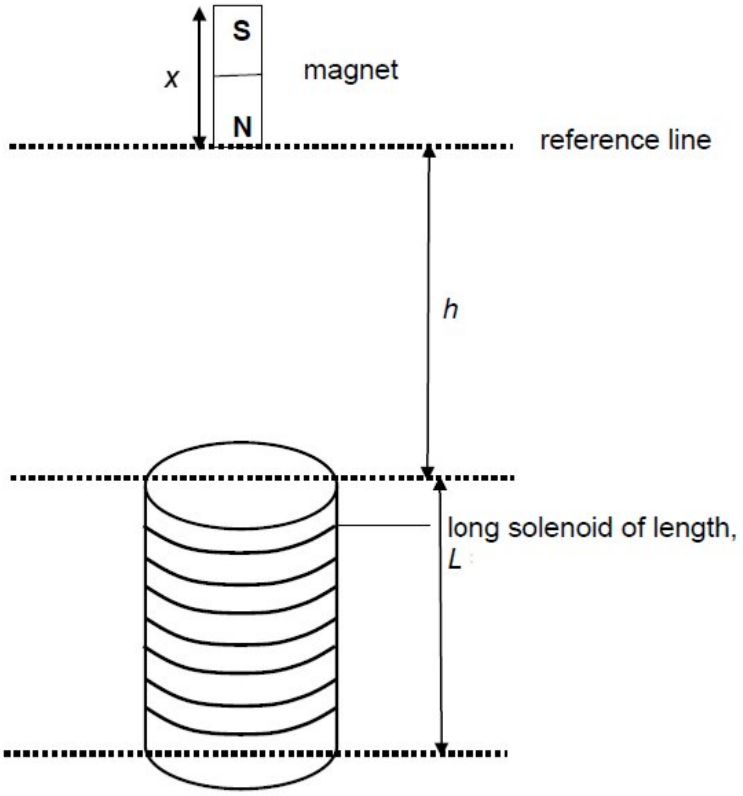


Fig. 5.2

(i) At time t = 0.0 s, the magnet is released from the reference line and allowed to pass through the entire length of the solenoid. Sketch, on Fig. 5.3, a graph to show the variation of the induced e.m.f E in the solenoid with time t. Label this graph as A. (Numerical values are not required and ignore the effects of air resistance.)



Fig. 5.3

[2]

(ii) State how your graph in **(c)(i)** would change if the same magnet is released with the south pole facing downwards instead.

.....

.....

.....[1]

(iii) The experiment is now repeated with the north pole of the magnet facing downwards, but released from a greater height above the solenoid. Sketch, on Fig. 5.3, a new graph to show the variation of the induced emf E in the solenoid with time t. Label this graph as B.

[2]

[Total: 10]